The `std::ratio_divide` template alias provides a mechanism for dividing two `std::ratio` values at compile time, using rational arithmetic.

Definition

```
template <class R1, class R2>
using ratio_divide = std::ratio<see below>;
```

Preconditions

`R1` and `R2` must be instantiations of the `std::ratio` class template.

Effects

`ratio_divide<R1,R2>` is defined as an alias for an instantiation of `std::ratio` that represents the result of dividing the fractions represented by `R1` and `R2` if that result can be calculated without overflow. If the calculation overflows, the program is ill formed. In the absence of arithmetic overflow, `std::ratio_divide<R1,R2>` shall have the same `num` and `den` values as `std::ratio<R1::num * R2::den, R1::den *R2::num>`.

Examples

```
std::ratio_divide<std::ratio<1,3>, std::ratio<2,5> >::num == 5
std::ratio_divide<std::ratio<1,3>, std::ratio<2,5> >::den == 6

std::ratio_divide<std::ratio<1,3>, std::ratio<15,7> >::num == 7
std::ratio_divide<std::ratio<1,3>, std::ratio<15,7> >::den == 45
```

D.6.6. std::ratio_equal class template

The `std::ratio_equal` class template provides a mechanism for comparing two `std::ratio` values for equality at compile time, using rational arithmetic.

Class definition